

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Analyzed structured and unstructured production data using **statistical analysis**, **data mining**, and **Python** (pandas, NumPy, scikit-learn) to identify root cause of yield and performance issues in enterprise manufacturing systems, extracting actionable insights that drove corrective action plans and improved process health by **18%**.
- Collaborated with cross-functional teams including Technology Development, Device Engineering, Defect Reduction, and Yield Analysis to own product maintenance workflows, implement continuous improvement procedures, and optimize targets for yield, device performance, and manufacturing quality across interconnected production systems.
- Assessed quality and reliability risks for manufacturing process changes and material updates by conducting **data-driven risk analysis** and statistical hypothesis testing, ensuring adherence to production standards and preventing yield excursions through rigorous process validation and change-control procedures.
- Established and implemented **structured workflows and procedures** for continuous improvement programs using **Jenkins CI/CD pipelines**, **Git version control**, and **automated data pipelines** to standardize quality checks, reduce manual defect review cycles by **35%**, and ensure consistent sort and line process execution.
- Collaborated with Technology Development and Local Yield teams to import new technology to production environments by developing **data integration workflows**, **ETL processes**, and **Power BI dashboards** that provided real-time process health visibility, enabling rapid decision-making on improvement activities and technology transition timelines.
- Managed high-pressure incident response and production support for mission-critical systems using **Splunk and CloudWatch monitoring**, **SQL performance tuning**, and **Linux shell scripting**, resolving P1/P2 defects within SLA windows and maintaining **99.2% system uptime** across manufacturing operations.

SKILLS

Process Engineering & Manufacturing Analytics – FEOL/BEOL process optimization, yield analysis, defect reduction, lithography and advanced patterning, dry etch processes, wet cleaning technologies, diffusion processes, thin-film deposition, CMP, device physics.

Data Mining & Statistical Analysis – hypothesis testing, regression analysis, exploratory data analysis, time-series forecasting, anomaly detection, power analysis, risk assessment, decision-making under uncertainty.

Data Engineering & BI – Python (pandas, NumPy, scikit-learn), Power BI dashboards and DAX, Tableau, SQL (PostgreSQL, MySQL, Oracle), ETL pipelines, data modelling, data warehousing, R/RStudio.

Manufacturing Systems & Integrations – ERP integration, cross-system workflows, REST API integrations, change-control management, UAT coordination, production support, SLA management, quality assurance.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker containerization, Jenkins CI/CD pipelines, Git version control, Linux, shell scripting, automated monitoring and alerting.

Communication & Collaboration – cross-functional team leadership, stakeholder communication, complex problem presentation, technical documentation, Agile/Scrum methodologies.

EDUCATION AND TRAINING

PROJECTS

Manufacturing Yield & Defect Reduction Intelligence Platform (HCL Technologies)

- Designed and deployed an integrated **data mining and statistical analysis platform** that ingested production defect data, process parameters, and device performance metrics from manufacturing equipment using **Python ETL pipelines** and **SQL-based data warehousing**, consolidating **2.5M+ daily sensor records** across FEOL and BEOL process nodes.
- Developed **Power BI dashboards** with **DAX measures** and interactive visualizations that enabled root-cause analysis of yield loss events, quality failures, and performance deviations in real time, reducing defect investigation cycles from **8 hours to 90 minutes** and accelerating corrective action deployment.
- Implemented **statistical hypothesis testing** and **regression modeling** workflows using **R and Python scikit-learn** to quantify the impact of manufacturing process changes (lithography, etch, deposition, CMP parameters) on yield and device reliability, validating **16 process improvements** that collectively improved manufacturing yield by **12.3%**.
- Established automated **continuous improvement procedures** using **Jenkins pipelines** and **scheduled Python scripts** to monitor process health metrics against control limits, trigger early-warning alerts for yield excursions, and generate weekly executive reports for Technology Development and Local Yield teams—reducing production incidents by **31%**.
- Collaborated with Device Engineering, Defect Reduction, and Quality Assurance teams to assess and quantify risks associated with manufacturing material substitutions and process recipe changes, conducting **statistical risk analysis** and ensuring adherence to Intel manufacturing standards before fab deployment.

Production Quality & Reliability Assurance System (HCL Technologies)

- Built an enterprise **production monitoring and incident management system** integrating **Splunk, CloudWatch, and Dynatrace** to provide real-time visibility into manufacturing process health, equipment performance, and quality metrics across multiple production fab lines, enabling rapid root-cause identification and SLA-compliant defect response.
- Developed **SQL-based data validation queries** and **automated quality checks** deployed via **Jenkins CI/CD pipelines** to verify manufacturing data integrity, detect anomalies in process parameters, and flag non-conforming product batches before sort and packaging, preventing **18+ quality escapes** annually.
- Established cross-functional collaboration workflows with manufacturing operations, quality assurance, and technology development teams to drive continuous improvement initiatives, coordinate change-control procedures for process modifications, and execute UAT validation for new equipment and process recipes, achieving **99.2% production uptime**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)